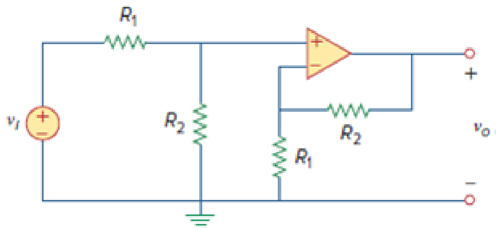
Chapter 5, Problem 29P

Problem

Determine the voltage gain v_o/v_i of the op amp circuit in Fig. 5.67.

Figure. 5.67:



Step-by-step solution

Step 1 of 3

Refer to Figure 5.67 in the textbook.

The modified circuit is shown in Figure 1.

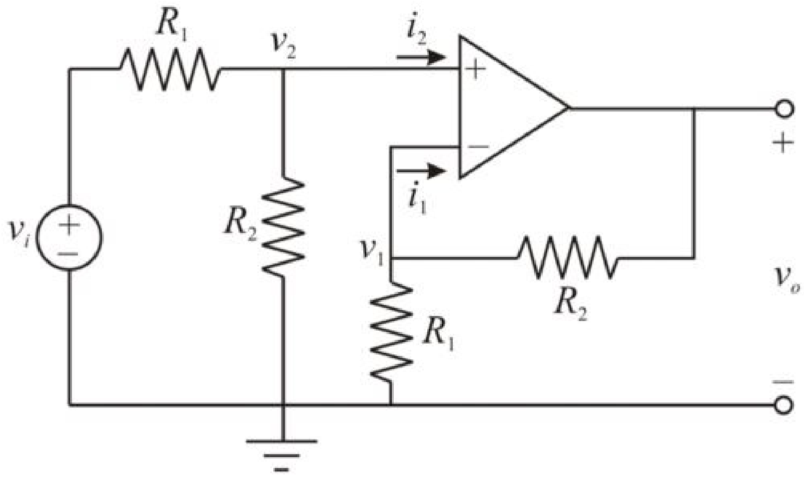


Figure 1

Step 2 of 3

In an ideal op-amp, the inverting and non-inverting terminal currents are zero because of infinite input impedance, thus,

$$i_1 = 0\,\mathrm{A}$$
$$i_2 = 0\,\mathrm{A}$$

The inverting and non-inverting node voltages are equal due to virtual short circuit, thus,

$v_1 = v_2$ ...... (1)

Applying Kirchhoff's current law at node v_2.

$$\frac{v_2 - v_i}{R_1} + \frac{v_2}{R_2} = 0$$
$$v_2\left[\frac{1}{R_1} + \frac{1}{R_2}\right] - \frac{v_i}{R_1} = 0$$
$$v_2\left[\frac{1}{R_1} + \frac{1}{R_2}\right] = \frac{v_i}{R_1}$$
$$v_2\left[\frac{R_1 + R_2}{R_1 R_2}\right] = \frac{v_i}{R_1}$$
$$v_2\left[\frac{R_1 + R_2}{R_2}\right] = v_i$$
$$v_2 = v_i\left[\frac{R_2}{R_1 + R_2}\right]$$

Since $v_1 = v_2$, therefore, substitute $v_i\left[\frac{R_2}{R_1 + R_2}\right]$ for v_2 in equation (1).

$v_1 = v_i\left[\frac{R_2}{R_1 + R_2}\right]$ ...... (2)

Step 3 of 3

Applying Kirchhoff's current law at node v_1.

$$\frac{v_1}{R_1} + \frac{v_1 - v_o}{R_2} = 0$$
$$v_1\left[\frac{1}{R_1} + \frac{1}{R_2}\right] - \frac{v_o}{R_2} = 0$$
$$v_1\left[\frac{1}{R_1} + \frac{1}{R_2}\right] = \frac{v_o}{R_2}$$
$$v_1\left[\frac{R_1 + R_2}{R_1 R_2}\right] = \frac{v_o}{R_2}$$

$v_o = v_1\left[\frac{R_1 + R_2}{R_1}\right]$ ...... (3)

Substitute $v_i\left[\frac{R_2}{R_1 + R_2}\right]$ for v_1 in equation (3).

$$v_o = v_i\left[\frac{R_2}{R_1 + R_2}\right]\left[\frac{R_1 + R_2}{R_1}\right]$$
$$v_o = v_i\left[\frac{R_2}{R_1}\right]$$
$$\frac{v_o}{v_i} = \frac{R_2}{R_1}$$

Hence, the voltage gain $\frac{v_o}{v_i}$ is $\boxed{\frac{R_2}{R_1}}$.